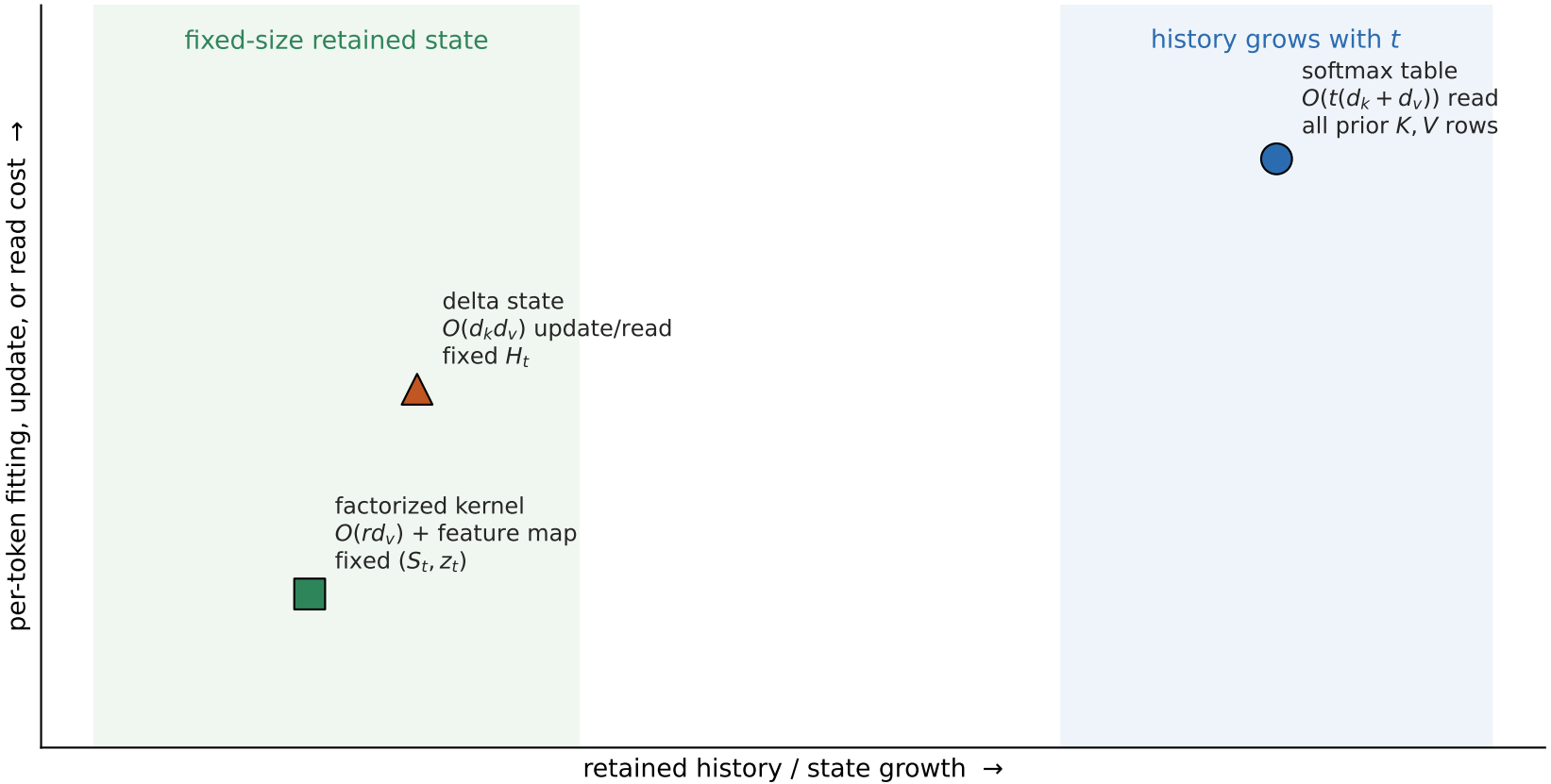
three memory constructions, three costs
fixed-size retained state
history grows with t
softmax table
$O(t(d_k + d_v))$ read
all prior K, V rows
delta state
$O(d_k d_v)$ update/read
fixed H_t
factorized kernel
$O(rd_v)$ + feature map
fixed (S_t, z_t)
per-token fitting, update, or read cost →
retained history / state growth →